03-621 Week 8

Advanced Quantative Genetics

Aidan Jan

March 9, 2026

Complex and Quantitative Traits

Importance of Complex Trait Analysis in Humans

1. Identification of genetic variations that act as disease risk factors:
   (a) Basis of tests to determine predisposition or susceptibility, allowing early diagnosis and thus more successful treatment
   (b) The identified genes (or their networks) may serve as targets for therapies
   (c) Precision Medicine: understanding the genetic heterogeneity of a disorder may allow treatments better tailored to the individual
2. Defining the genetic variables simplifies analysis of the environmental risk factors.

Essential Features of Complex Traits

The phenotypes can be described quantitatively. Each phenotype of a complex trait can be assigned:

- a quantitative value x
- a frequency y in a population

Each trait exhibits a **distribution** of "phenotypic values" in a population

Three Types of Complex Traits

Continuous Traits

- Vary continuously from one phenotypic extreme to the other
- Cumulative small effects by many genes, plus environment
- Example: height, weight, hand size, skin color, etc.

Meristic Traits

- Phenotype corresponds to one of several **discrete** categories
- Additive effects of alleles at multiple genes
- Example: Number of scales on a fish, number of offspring at a time

Threshold Traits

- Only two phenotypic classes ("dichotomous")
- Each individual has an underlying **total liability** ("risk") to express the trait.
- Individuals whose total risk (genes + environment) exceeds a threshold value will express the trait.
- Example: whether or not an individual has schizophrenia

Complex Traits exhibit "Non-Mendelian" Inheritance

- The same genotype can yield different phenotypes (depending environments)
- Many different genotypes can yield the same phenotypes
- They cannoy be followed easily through pedigrees
- It can be difficult to determine that they are heritable at all.

If inheritance is particulate (per Mendel), why do we get this behavior for most traits?

- Continuous phenotype distributions
- and "regression towards the mean": on average, parents of different heights have children of intermediate height

Distributions of Complex Trait phenotypes can usually be approximated by a Gaussian Distribution (aka Normal)

- Populations can differ in Mean Phenotypic Value and Phentypic Variance for any given complex trait

Complex Traits Summary

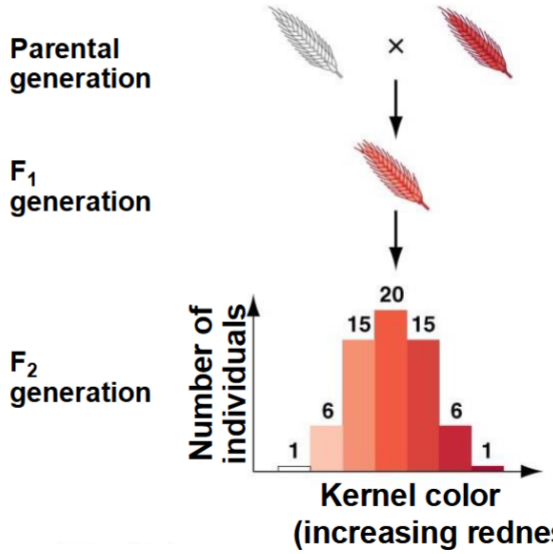
- What are complex traits?
  - Multiple genetic and environmental factors
  - Complex inheritance
- Three types of complex traits
  - Continuous: height, weight, blood pressure
  - Meristic: # puppies per litter
  - Threshold: schizophrenia, diabetes
- Distributions of phenotypes approximated by a Normal distribution

Quantitative Traits

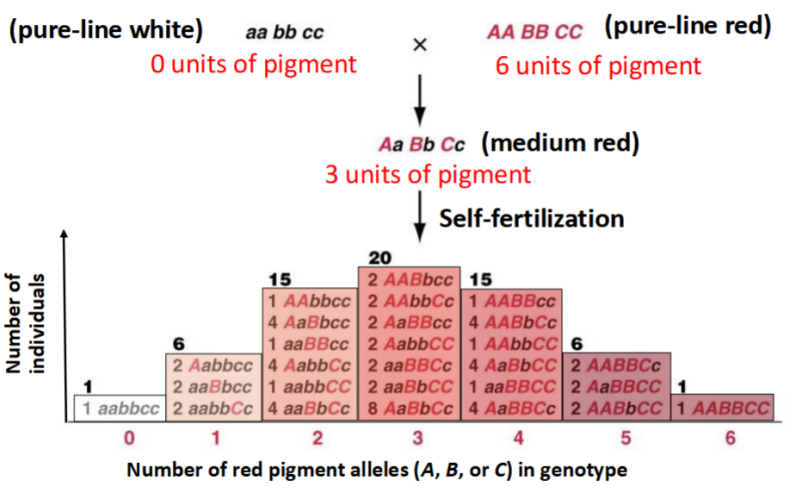
Quantitative Traits are Based on Polygenic Inheritance

Determined by the combined actions of many genes:

- Alleles at each gene contribute a small (additive) amount to the differences in phenotypic value
- Thus, the population usually exhibits a Gaussian distribution of phenotype values
- For example, wheat kernel color is a quantitative trait.



7 color phenotypes (under controlled environment)



Mean
$= [\Sigma(f_i x_i)]/64 = 3$

Variance
$= [\Sigma f_i(x_i - 3)^2]/64 = 1.5$

Nilson-Ehle 1908

- The distribution of genotypes corresponds to the 7 color phenotypes.
- Superimposed environmental effects can smooth out the distribution even more

Polygenic Inheritance can Give Rise to Fine-Grained Phenotypic Variation

With just two alleles per gene: # genotypes $= 3^n$.

- There are three since (homozygous dominant, homozygous recessive, and heterozygous)

Relative Importance of Genotype and Environment in Complex Traits

- Can we use phenotypic mean? Why?
- How about genetic versus environmental contributions to the variance in phenotype?